



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar - 4

Deep Learning in Computer Vision with Sensors: Application in Assistance for Visually Impaired People

DATE & TIME:
27th January 2021,
4:00 PM.



SPEAKER

Rakesh Chandra Joshi

Research Scholar,
Centre for Advanced Studies
AKTU, Lucknow

PLATFORM & WEB-LINK
Google Meet :
<https://meet.google.com/pxa-emzj-xka>



ABSTRACT

Deep learning (DL) allows computational models of multiple processing layers to learn and represent data with multiple levels of abstraction mimicking how the brain perceives and understands multimodal information, thus implicitly capturing intricate structures of large-scale data. DL methods improve prediction performance using big data and plentiful computing resources. Deep learning is a rich family of methods, encompassing neural networks, hierarchical probabilistic models, and a variety of unsupervised and supervised feature learning algorithms. Problems which were assumed to be unsolvable are now being solved with super-human accuracy. Visually impaired people who face numerous difficulties in their daily life, an artificial intelligence-based fully automatic assistive framework is discussed to recognize different objects, and auditory inputs are provided to the user in real-time, which gives better understanding about their surroundings. In addition to computer vision-based techniques for object recognition, sensors are integrated to make the device more comprehensive by recognizing obstacles while navigating from one place to another with low time complexity.